INCH-POUND
MIL-DTL-24643/24B
22 August 2002
SUPERSEDING
MIL-C-24643/24A
14 March 1994

DETAIL SPECIFICATION SHEET

# CABLE, ELECTRICAL, TYPES LSTCJX, LSTCKX AND LSTCTX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

| | |
|---|---|
| First - | Stranded conductor, uncoated (see table I for size and stranding). |
| Second - | Extruded silicone rubber insulation (see table I for dimensions). |
| Third - | Glass braid, color identified (see table II). (see table I for dimensions). |
| Fourth - | Transparent braid covering. Two conductors cabled together to form a pair with lay not greater than 2.25 inches. For LSTCJX, one iron conductor and one constantan conductor. For type LSTCKX, one chromel and one alumel conductor. For LSTCTX, one copper conductor and one constantan conductor. Type J, K and T thermocouple conductors shall be in accordance with ANSI MC96.1. Color code applied by method 4, see table II for color code sequence. |
| Fifth - | The specified number of pairs (see table I) cabled together with a lay not greater than 10 inches. Fillers saturated with non-charring silicone base compound to form a firm, well-rounded assembly. |
| Sixth - | A binder tape applied helically with overlap. |
| Seventh - | Silicone rubber jacket colored orange or red. (see table I for thickness) |
| Eighth - | Braided metal armor. |



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TABLE I. Details.

| Military part no. M24643/24 | Type and size | Conductors | | | | Cable jacket thickness min avg (in) | Overall dia max (in) | Cold bending mandrel dia max (in) |
|---|---|---|---|---|---|---|---|---|
| | | No. of pairs | Number of strands and strand diameter (inch) | Diameter over insulation nominal (inch) | Diameter over braid covering maximum (inch) | | | |
| -01AN | LSTCJX-3 | 3 | 7/.0201 | 0.100 | 0.125 | 0.050 | 0.742 | 9.0 |
| -02AN | LSTCJX-7 | 7 | 7/.0201 | 0.100 | 0.125 | 0.062 | 0.983 | 11.0 |
| -03AN | LSTCJX-12 | 12 | 7/.0201 | 0.100 | 0.125 | 0.062 | 1.269 | 15.0 |
| -04AN | LSTCKX-1 | 1 | 7/.0201 | 0.100 | 0.125 | 0.038 | 0.456 | 5.0 |
| -05AN | LSTCKX-3 | 3 | 7/.0201 | 0.100 | 0.125 | 0.050 | 0.742 | 9.0 |
| -06AN | LSTCKX-7 | 7 | 7/.0201 | 0.100 | 0.125 | 0.062 | 0.983 | 11.0 |
| -07AN | LSTCKX-12 | 12 | 7/.0201 | 0.100 | 0.125 | 0.062 | 1.269 | 15.0 |
| -08AN | LSTCTX-1 | 1 | 7/.0113 | 0.065 | 0.085 | 0.038 | 0.350 | 4.25 |
| -09AN | LSTCTX-3 | 3 | 7/.0113 | 0.065 | 0.085 | 0.038 | 0.552 | 6.5 |
| -10AN | LSTCTX-7 | 7 | 7/.0113 | 0.065 | 0.085 | 0.050 | 0.731 | 8.0 |
| -11AN | LSTCTX-12 | 12 | 7/.0113 | 0.065 | 0.085 | 0.063 | 0.964 | 11.0 |

TABLE II. Color identification code.

| Pair number | Constantan or alumel | Iron, copper or chromel pair | |
|---|---|---|---|
| | | Base color | Tracer color |
| 1 | Red | White | ---- |
| 2 | Red | Black | ---- |
| 3 | Red | Green | ---- |
| 4 | Red | Orange | ---- |
| 5 | Red | Blue | ---- |
| 6 | Red | White | Black |
| 7 | Red | White | Green |
| 8 | Red | White | Orange |
| 9 | Red | White | Blue |
| 10 | Red | White | Red |
| 11 | Red | Black | White |
| 12 | Red | Green | White |

EXAMINATION AND TESTS:

Basic electrical: Requirements:

Voltage withstand - volts, root mean square, minimum .....2000

Insulation resistance - megohms -1000 feet, minimum

Conductor to conductor .....500



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ANSI MC96.1 calibration limits of error:
(a) types LSTCJX ..... ± 2.2°C
(b) types LSTCKX, minus 32 to 530°F ..... ± 4°C
532 to 2300°F.....
(c) Types LSTCTX ..... ± 1°C
Conductor continuity ..... No failure

Group A:
Visual and dimensional..... No failure
Watertightness - see MIL-DTL-24643 for limits of water leakage ..... No failure
Group B:
Cross-linked proof test (percent, maximum)
Insulation .....50
Jacket (when tested at 200 °C).....50
Gas flame - 1 hour at 120/70 volt root mean square..... No failure
Cold bending, cable - (see Table I for mandrel diameter) ..... No damage
Physicals (unaged)
Insulation (extruded)
Tensile strength - lb/in$^2$, minimum .....750
Elongation - percent, minimum .....125
Jacket (cable)
Tensile strength - lb/in$^2$, minimum .....800
Elongation - percent, minimum .....250
Durometer (type A) hardness, minimum (ASTM D 2240).....60
Group C:
Physicals - cable aging (260 ± 10°C)
Jacket
Tensile strength - lb/in$^2$, minimum .....600
Elongation - percent, minimum .....150
Group D:
Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Requirements:
Gas flame (3 hours) at 120/70 volts RMS ..... No failure
Aging and compatibility (cable)(125 ± 5°C) ..... No failure
For this test, omit all references to watertightness and fillers.
Armor - conformance to material, construction and coverage ..... No failure
Specific gravity of extruded silicone insulation, maximum .....1.55
Halogen content - percent, maximum
Jacket .....0.2
Fillers .....0.2



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Smoke index, maximum
- Jacket .....35
- Fillers .....45
- Insulation .....35

Toxicity index, maximum
- Jacket .....5
- Fillers .....5
- Insulation .....1.5

UNIT ORDERING LENGTHS: All types and sizes 100 feet minimum.

NOTE: Not for AIR FORCE use.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-022)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC



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